

# Sujith S

System Development Engineer | Python · Linux/Unix · AWS · Infrastructure Automation · Production Systems  
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## Summary

Computational Statistics and Data Analytics student at VIT University (2022-2027) with hands-on experience developing automation and operating complex systems in Linux-based production environments. Proficient in Python and Java; applied experience in infrastructure automation, CI/CD pipelines, Linux-based Docker deployments, distributed systems networking (Kafka, REST/HTTP), and production system operations on AWS. Proven ability to triage operational incidents quickly, build reliable and correct automation, and collaborate closely with engineering teams to improve system stability, correctness, and reduce manual operational overhead.

## Education

### Int.MSc Computational Statistics and Data Analytics

VIT University | 2022-2027

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Distributed Systems, Cloud Computing, Computer Networks, Database Systems, Operating Systems, Machine Learning

### CS50 Programs

Harvard CS50 (Online)

Completed [CS50x](#), [CS50P](#), [CS50AI](#), [CS50SQL](#), and [CS50R](#) covering algorithms, systems programming, SQL, Python, and full-stack development.

## Skills

**Languages:** Python, Java, SQL, JavaScript, TypeScript, R

**Linux & Systems:** Linux/Unix, Shell Scripting, Docker, System Administration, Log Analysis, Process Management

**Networking & Distributed Systems:** TCP/IP, REST/HTTP, Apache Kafka (broker/consumer networking), Apache Spark, DNS, API Gateway

**Cloud & Automation:** AWS (Lambda, S3, RDS, EKS, EventBridge), GCP, Azure, CI/CD Pipelines, Infrastructure Automation, Build/Release/Deployment Automation

**Operations & Monitoring:** System Monitoring, Triage & Incident Response, Root Cause Analysis, PostgreSQL, Redis, MLflow, Troubleshooting

## Projects

### Social Media Sentiment Lakehouse

[GitHub](#)

*Python, Apache Kafka, Apache Spark Streaming, Delta Lake, FastAPI, PostgreSQL, Docker, Linux, AWS*

- Deployed and operated an end-to-end real-time streaming pipeline on Linux-based AWS infrastructure using Apache Kafka (multi-broker networking, producer/consumer configuration) and Apache Spark Streaming; managed Docker containerized service operations, ensuring reliable high-throughput data processing and operational stability across distributed components.
- Automated monitoring, failure detection, and system recovery workflows using Python scripting; triaged production incidents through log analysis and process inspection; implemented alerting pipelines to reduce mean time to recovery and eliminate manual operational intervention.

### Demand Forecasting System

[GitHub](#)

*Python, FastAPI, PostgreSQL, Redis, MLflow, Docker, Linux, AWS*

- Designed, deployed, and operated production REST APIs on Linux-based AWS infrastructure using FastAPI and Docker; configured PostgreSQL and Redis for high-concurrency request handling; implemented staging-to-production deployment pipelines with automated validation and rollback to ensure reliable, correct system rollout.
- Developed automated build, test, and release pipelines to eliminate manual deployment steps; instrumented drift detection and system health monitoring for production reliability; tracked operational metrics and experiment reproducibility using MLflow.

## Autonomous Research Agent

[GitHub](#)

*Python, LangChain, LangGraph, FastAPI, FAISS, PostgreSQL, Redis, Docker, Linux, AWS*

- Architected and operated a distributed multi-service system on Linux-based AWS infrastructure using FastAPI, PostgreSQL, Redis, and Docker; managed inter-service networking and REST/HTTP communication; owned component design, implementation, testing, and production deployment end-to-end.
- Automated system health checks, evaluation pipelines, and operational benchmarking using Python scripting; documented system runbooks and operational procedures; collaborated with engineers to triage service failures, identify root causes, and resolve defects affecting system correctness.

## Work Experience

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### Founder

[Shoot the Breeze](#) | 2023 - Present

- Owned end-to-end SDLC for production applications deployed on Linux-based AWS infrastructure: designed system architecture, implemented automation in Python, managed Docker container operations across staging and production, and built automated monitoring and incident triage workflows to ensure system reliability, correctness, and rapid defect resolution.

### App/Web Developer

[CropSky](#) | VIT-TBI Research & Innovation Hub | 2024 - Present

- Developed and deployed scalable REST APIs and backend systems on Linux-based cloud infrastructure at a university-incubated startup; implemented CI/CD pipelines and automated build and release processes; collaborated cross-functionally with engineers to triage operational issues, identify root causes, and deliver reliable, production-ready system components.

## Certifications & Achievements

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- [Google Generative AI Leader](#)
- [Google Cloud Computing Foundations](#)
- [FreeCodeCamp Legacy Full Stack](#)
- [OCI 2025 Certified Developer Professional](#)
- AWS [Solutions Architect](#) and [Developer](#)
- [Software Engineer Certification - HackerRank](#)
- Published a [research paper](#) on [stock market analysis and prediction](#) in the 14<sup>th</sup> proceedings of the [El Ruha International Conference on Scientific Research](#), pp. 730-740, 2025.